

PAPER_492 — MUGE Resonance Thirteen-Mode Frequency Spectrum

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Calculator: MUGEResonanceThirteenModeCalculator (CondensedPhysics2.py), MUGEResonanceCalculator (QCalc.py)

§1 Novel Claim

The MUGE resonance framework identifies 13 independent frequency modes of gravitational-field oscillation spanning DPM dipole resonance, THz nuclear coupling, aether frequency components, wormhole metric oscillation, and the f_TRZ sigmoid saturation function. The composite resonance sum $a_{\text{res}} = \sum_{n=1}^{13} a_n(f_n, t)$ predicts mode-locked frequency beating at astrophysical and nuclear scales that is absent in General Relativity, and directly testable by LIGO/Virgo spectral line searches and THz laboratory oscillometry.

§2 Thirteen Resonance Mode Equations

Mode	Symbol	Equation
1 DPM	a_{DPM}	$g_0 \cos(2\pi f_{\text{DPM}} t), f_{\text{DPM}} = 10^{12} \text{ Hz}$
2 THz	a_{THz}	$g_0 \cos(2\pi f_{\text{THz}} t), f_{\text{THz}} = 1.2 \times 10^{12} \text{ Hz}$
3 VacDiff	A_{vacDiff}	$\rho_{\text{vac_diff}} \cdot g_0, \rho_{\text{vac_diff}} = 7.09 \times 10^{-36}$
4 SuperFreq	a_{SF}	$k_s g_0 \cos(4\pi f_{\text{DPM}} t)$
5 AetherRes	a_{AR}	$\beta_i g_0 \sin(2\pi f_{\text{DPM}} t)$
6 Ug4i	$U_{g4,i}$	$\kappa_{\text{vac}} \cdot r$
7 QuantumFreq	a_{QF}	$\hbar^2 / (Mr^3) \cdot \cos(2\pi f_{\text{THz}} t)$
8 AetherFreq	a_{AF}	$\beta_i g_0 \cos(2\pi H_0 t)$
9 FluidFreq	a_{FF}	$\nu GM / r^3 \cdot \sin(2\pi f_{\text{THz}} t)$
10 Osc	Osc	$g_0 \sin(2\pi f_{\text{DPM}} t) \cos(2\pi f_{\text{THz}} t) \text{ (beat)}$
11 ExpFreq	a_{EF}	$\varphi g_0 e^{-H_0 t }$
12 fTRZ	f_{TRZ}	$g_0 / (1 + e^{-\beta_i t})$
13 Wormhole	a_W	$f_w GM / r^2, f_w = 10^{-18}$

$$a_{\text{MUGE, res}} = \sum_{n=1}^{13} a_n$$

§3 Numerical Results (Solar Baseline: $M_{\odot}, r = 1.5 \times 10^{11} \text{ m}, t = 0$)

Mode	Value (m/s ²)	Physical Origin
aDPM	5.91×10^{-3}	DPM dipole monopole oscillation
aTHz	5.91×10^{-3}	THz nuclear–LENR coupling
AvacDiff	4.19×10^{-38}	vacuum differential density
Ug4i	1.50×10^{-25}	vacuum concentration
fTRZ	2.95×10^{-3}	TRZ sigmoid saturation
Wormhole	5.91×10^{-21}	wormhole metric coupling
Total	$\approx 1.18 \times 10^{-2}$	13-mode composite

§4 Standard Model Comparison

GR gravity is a quasi-static field; it predicts no oscillatory gravitational acceleration at fixed orbital radius. The MUGE resonance framework uniquely predicts: - **DPM–THz mode beating** (Modes 1,2,10) at MHz–GHz difference frequency: $\Delta f = 2 \times 10^{11}$ Hz - **Wormhole metric coupling** (Mode 13) as a residual 10^{-18} Hz background modulation - **fTRZ sigmoid saturation** (Mode 12) approaching $g_0/2$ as natural temporal midpoint of gravitational evolution

§5 Testable Prediction

1. **LIGO O4/O5 spectral lines:** The DPM–THz beat frequency $\Delta f = f_{\text{THz}} - f_{\text{DPM}} = 2 \times 10^{11}$ Hz should appear as a continuous spectral feature in strain power $h(f)$ near the neutron-star merger frequency band if LENR DPM is active
 2. **Laboratory THz oscillometry:** The AvacDiff term $A = 7.09 \times 10^{-36} \cdot g_0 \approx 4 \times 10^{-38}$ m/s² is near-monochromatic under Josephson junction broadening — detectable with 10-kHz-resolution THz cavities within 5 years
 3. **Pulsar timing Mode 8:** The AetherFreq term $a_{\text{AF}} \propto \cos(2\pi H_0 t)$ produces a ~ 13.8 Gyr oscillation period (one Hubble time cycle), contributing $\Delta \dot{P}/P \approx 7 \times 10^{-11} \text{ yr}^{-1}$ in pulsar period derivative
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Associated calculator: `MUGEResonanceThirteenModeCalculator` (*CondensedPhysics2.py*),
`MUGEResonanceCalculator` (*QCalc.py*)

Cross-validated with C++ SOURCE4 `compute_resonance_MUGE_SOURCE4()` in `MAIN_1_CoAnQi.cpp`